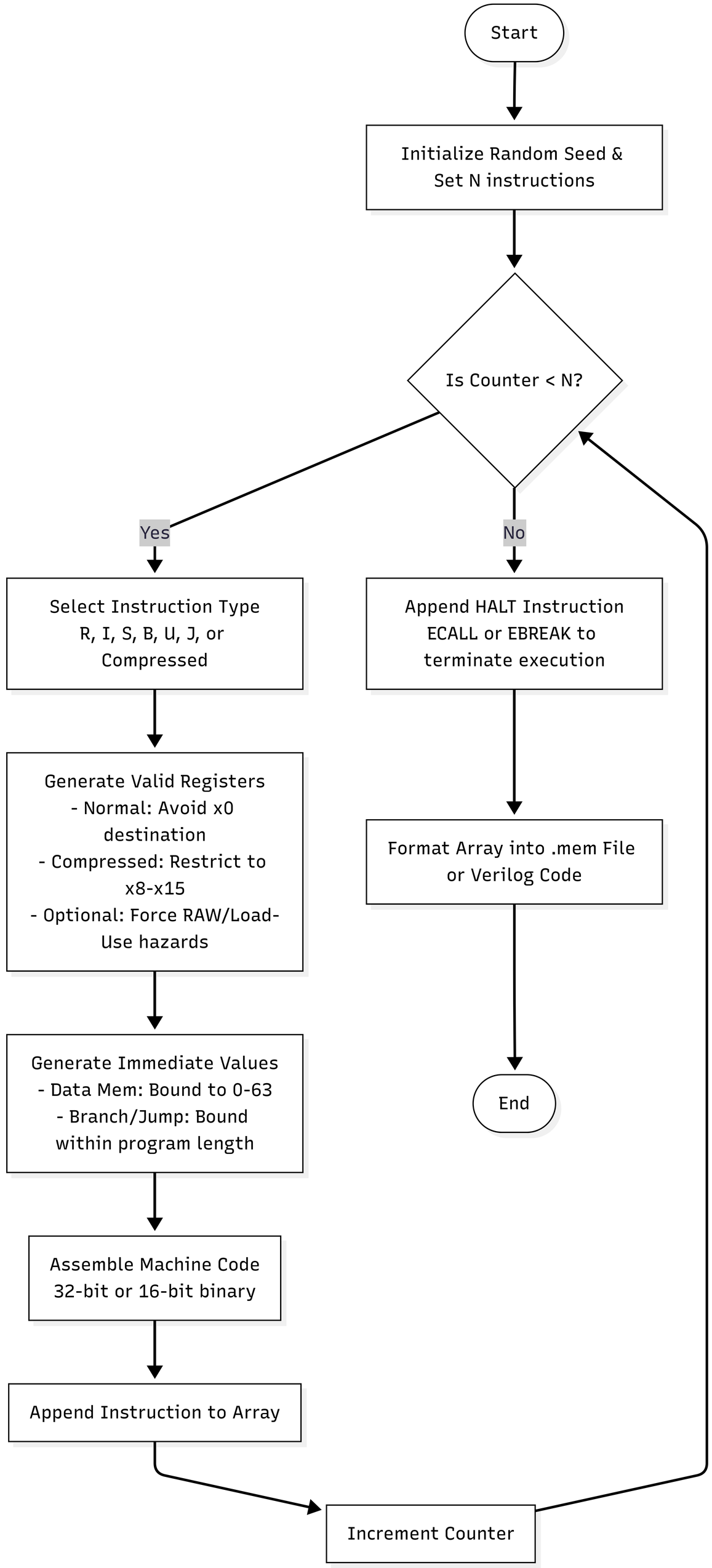
Start
Initialize Random Seed & Set N instructions
Is Counter < N?
Yes
No
Select Instruction Type R, I, S, B, U, J, or Compressed
Generate Valid Registers - Normal: Avoid x0 destination - Compressed: Restrict to x8-x15 - Optional: Force RAW/Load-Use hazards
Generate Immediate Values - Data Mem: Bound to 0-63 - Branch/Jump: Bound within program length
Assemble Machine Code 32-bit or 16-bit binary
Append Instruction to Array
Increment Counter
Append HALT Instruction ECALL or EBREAK to terminate execution
Format Array into .mem File or Verilog Code
End